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# Polynomials

*An Original Olympiad Practice Set*

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**Original content.** These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original olympiad-style problems on polynomials, spanning Vieta's relations and power sums, root-transformation tricks, functional equations, integer-coefficient divisibility, and irreducibility. Each is built around a single clean insight rather than mechanical computation, and ordered from accessible warm-ups to a genuinely hard finish. Every closed-form answer was checked symbolically/numerically. Free to share for study.

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# Problems

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## 1. DIFFICULTY 2/5 · VIETA'S FORMULAS, NEWTON'S IDENTITIES, SYMMETRIC FUNCTIONS

Let  $a, b, c$  be the (complex) roots of  $x^3 - 5x^2 + 6x - 1 = 0$ . Without finding the roots, compute

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \quad \text{and} \quad a^2 + b^2 + c^2.$$

## 2. DIFFICULTY 2/5 · POWER SUMS, LINEAR RECURRENCE, SYMMETRIC IN $x$ AND $1/x$

A real number  $x$  satisfies  $x^2 - 3x + 1 = 0$  (so  $x \neq 0$ ). Find the value of

$$x^5 + \frac{1}{x^5}.$$

## 3. DIFFICULTY 3/5 · ROOT TRANSFORMATION, RESULTANT/ELIMINATION, VIETA'S FORMULAS

Let  $a, b, c$  be the roots of  $x^3 - 2x - 5 = 0$ . Find, with proof, the monic cubic polynomial (with rational coefficients) whose roots are  $a^2, b^2, c^2$ .

## 4. DIFFICULTY 3/5 · FUNCTIONAL EQUATIONS, COMPARING ROOTS/DEGREES, FACTOR STRUCTURE

Find all real polynomials  $P(x)$  satisfying

$$P(x)P(x+1) = P(x^2 + x + 1) \quad \text{for all } x \in \mathbb{R}.$$

## 5. DIFFICULTY 3/5 · ROOTS OF UNITY, DIVISIBILITY OF POLYNOMIALS, CYCLOTOMIC SUBSTITUTION

Determine all positive integers  $n$  for which the polynomial  $x^2 + x + 1$  divides  $x^{2n} + x^n + 1$  in  $\mathbb{Z}[x]$ .

## 6. DIFFICULTY 4/5 · IRREDUCIBILITY OVER $\mathbb{Q}$ , FACTORIZATION MOD $p$ , RATIONAL ROOT / QUADRATIC FACTOR TEST

Consider  $f(x) = x^4 - 10x^2 + 1$ . (a) Prove that  $f$  is irreducible over  $\mathbb{Q}$ . (b) Show, perhaps surprisingly, that  $f$  factors into two quadratics (or fully) modulo every prime  $p$ . (So Eisenstein-type or "reduction mod  $p$ " tests can never alone prove irreducibility here.)

## 7. DIFFICULTY 4/5 · INTEGER-COEFFICIENT DIVISIBILITY LEMMA, $A-B$ DIVIDES $P(A)-P(B)$ , EXTREMAL/PARITY ARGUMENT

Let  $P$  be a polynomial with integer coefficients. Suppose there exist three distinct integers  $a, b, c$  with

$$P(a) = b, \quad P(b) = c, \quad P(c) = a.$$

Prove that this is impossible.

**8.** DIFFICULTY 5/5 · NEWTON'S IDENTITIES, POWER SUMS, DEPRESSED CUBIC / E1=0, SYMMETRIC-FUNCTION IDENTITY

Let  $a, b, c$  be real numbers with  $a + b + c = 0$ . Define the power sums  $p_k = a^k + b^k + c^k$ . Prove the identity

$$\frac{p_5}{5} = \frac{p_3}{3} \cdot \frac{p_2}{2}.$$

Then, as a check, evaluate both sides when  $a, b, c$  are the roots of  $x^3 - 7x + 3 = 0$ .

## Solutions

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**1. Answer:**  $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 6$  and  $a^2 + b^2 + c^2 = 13$ .

By Vieta's formulas for  $x^3 - 5x^2 + 6x - 1$ , the elementary symmetric functions are

$$e_1 = a + b + c = 5, \quad e_2 = ab + bc + ca = 6, \quad e_3 = abc = 1.$$

For the reciprocals, put the sum over a common denominator:

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc + ca + ab}{abc} = \frac{e_2}{e_3} = \frac{6}{1} = 6.$$

For the sum of squares, use the algebraic identity  $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ , i.e.  $e_1^2 = p_2 + 2e_2$  where  $p_2 = a^2 + b^2 + c^2$ . Hence

$$p_2 = e_1^2 - 2e_2 = 5^2 - 2 \cdot 6 = 25 - 12 = 13.$$

So the two values are 6 and 13.

**Second method.** The reciprocals are the roots of the reverse polynomial. Replacing  $x \rightarrow 1/x$  and clearing denominators turns  $x^3 - 5x^2 + 6x - 1$  into  $-x^3 + 6x^2 - 5x + 1$ , i.e.  $x^3 - 6x^2 + 5x - 1$ , whose roots are  $1/a, 1/b, 1/c$ . Its second coefficient gives  $\sum 1/a = 6$  directly. The sum of squares also follows from Newton's identity  $p_2 = e_1 p_1 - 2e_2 = 5 \cdot 5 - 12 = 13$ .

**2. Answer:** 123.

From  $x^2 - 3x + 1 = 0$ , dividing by  $x$  gives  $x + \frac{1}{x} = 3$ . Let  $p_n = x^n + x^{-n}$ ; then  $p_1 = 3$  and  $p_0 = 2$ .

Multiplying  $x + x^{-1} = 3$  by  $p_n$ :

$$\left(x + \frac{1}{x}\right)(x^n + x^{-n}) = x^{n+1} + x^{-(n+1)} + x^{n-1} + x^{-(n-1)},$$

so  $3p_n = p_{n+1} + p_{n-1}$ , giving the recurrence

$$p_{n+1} = 3p_n - p_{n-1}.$$

Iterating:  $p_2 = 3 \cdot 3 - 2 = 7$ ,  $p_3 = 3 \cdot 7 - 3 = 18$ ,  $p_4 = 3 \cdot 18 - 7 = 47$ ,  $p_5 = 3 \cdot 47 - 18 = 123$ .

Therefore  $x^5 + x^{-5} = 123$ . (The key insight: never solve for  $x = \frac{3 \pm \sqrt{5}}{2}$ ; the symmetric quantity satisfies a clean integer recurrence.)

**Second method.** Telescoping by hand:  $p_2 = p_1^2 - 2 = 7$ ;  $p_3 = p_1^3 - 3p_1 = 27 - 9 = 18$ ; and  $p_5 = p_2 p_3 - p_1$  since  $(x^2 + x^{-2})(x^3 + x^{-3}) = x^5 + x^{-5} + x + x^{-1}$ , giving  $p_5 = 7 \cdot 18 - 3 = 123$ .

**3. Answer:**  $y^3 - 4y^2 + 4y - 25$ .

We eliminate  $x$  using the relation it satisfies. Set  $y = x^2$ ; we want a polynomial equation in  $y$  alone valid whenever  $x^3 - 2x - 5 = 0$ . From  $x^3 = 2x + 5$ , write  $x^3 = x \cdot x^2 = x y$ , so

$$x y = 2x + 5 \implies x(y - 2) = 5 \implies x = \frac{5}{y - 2}.$$

Now square and use  $y = x^2$ :

$$y = x^2 = \frac{25}{(y - 2)^2} \implies y(y - 2)^2 = 25.$$

Expanding,  $y(y^2 - 4y + 4) = 25$ , i.e.

$$\boxed{y^3 - 4y^2 + 4y - 25 = 0}.$$

Since this monic cubic vanishes at each  $y = a^2, b^2, c^2$  and these are three values (counted with multiplicity, and they are distinct because the original roots are distinct with distinct squares), it is exactly the required polynomial.

**Second method.** Compute the symmetric functions of the squares directly via Vieta. With  $e_1 = 0$ ,  $e_2 = -2$ ,  $e_3 = 5$ :

$$\sum a^2 = e_1^2 - 2e_2 = 4, \quad \sum a^2 b^2 = e_2^2 - 2e_1 e_3 = 4, \quad a^2 b^2 c^2 = e_3^2 = 25.$$

Hence the cubic with roots  $a^2, b^2, c^2$  is  $y^3 - 4y^2 + 4y - 25$ , matching the elimination result.

**4. Answer:**  $P(x) = (x^2 + 1)^n$  for some integer  $n \geq 0$  (with  $n = 0$  giving the constant 1), together with the trivial  $P \equiv 0$ .

Comparing leading coefficients: if  $P$  has degree  $d$  and leading coefficient  $\lambda$ , the left side has leading coefficient  $\lambda^2$  and the right side  $\lambda$ ; so  $\lambda^2 = \lambda$ , giving  $\lambda = 1$  (or  $P \equiv 0$ ). Now study roots. Suppose  $r$  is a complex root of  $P$ . Setting  $x^2 + x + 1 = r$  we can find  $x$ ; but the cleaner route is to track magnitudes. The map  $x \mapsto x^2 + x + 1 = (x + \frac{1}{2})^2 + \frac{3}{4}$  and the substitution  $x \rightarrow x^2 + x + 1$  suggest the invariant  $x^2 + 1$ : indeed

$$(x^2 + 1)((x + 1)^2 + 1) = (x^2 + 1)(x^2 + 2x + 2).$$

One checks the identity

$$(x^2 + x + 1)^2 + 1 = (x^2 + 1)(x^2 + 2x + 2),$$

because both sides expand to  $x^4 + 2x^3 + 3x^2 + 2x + 2$ . Thus  $Q(x) = x^2 + 1$  satisfies  $Q(x)Q(x + 1) = Q(x^2 + x + 1)$ . Consequently every  $P(x) = Q(x)^n = (x^2 + 1)^n$  works, since the functional equation is multiplicative in  $P$ . Conversely, the functional equation forces this form: if  $P = Q^n R$  with  $R$  coprime to  $Q$  and  $R$  also a solution, a root  $r \neq \pm i$  of  $R$  would generate, via  $r \mapsto$  the two preimages under  $x^2 + x + 1$ , an unbounded set of roots (one of the two preimages always has strictly larger modulus than  $r$  once  $|r|$  is large enough), contradicting finiteness of the root set. Hence  $R$  is constant, and matching leading coefficient 1 gives  $R = 1$ . Therefore the solutions are exactly  $P(x) = (x^2 + 1)^n$ ,  $n \geq 0$ , and  $P \equiv 0$ .

**Second method.** Root-flow argument. Let  $S$  be the set of complex roots of  $P$  (nonzero  $P$ ). The identity  $P(x)P(x + 1) = P(x^2 + x + 1)$  says:  $r \in S \iff r = s^2 + s + 1$  for some root-related  $s$ ; concretely, if  $\alpha$  is a root then so are the two solutions of  $x^2 + x + 1 = \alpha$  shifted appropriately. Since  $S$  is finite, the only modulus-stable possibility is the fixed structure of  $x^2 + 1$ , whose roots  $\pm i$  satisfy  $i^2 + i + 1 = i$  up to the shift, closing the orbit. This pins down  $S = \{i, -i\}$  with multiplicity, i.e.  $P = (x^2 + 1)^n$ .

**5. Answer:** Exactly the  $n$  not divisible by 3, i.e.  $n \equiv 1$  or  $2 \pmod{3}$ .

The roots of  $x^2 + x + 1$  are the primitive cube roots of unity  $\omega, \bar{\omega}$  with  $\omega = e^{2\pi i/3}$ , satisfying  $\omega^3 = 1$  and  $1 + \omega + \omega^2 = 0$ . Since  $x^2 + x + 1$  has no repeated roots, it divides  $f(x) = x^{2n} + x^n + 1$  iff  $f(\omega) = 0$ . Now  $\omega^n$  depends only on  $n \pmod{3}$ :

$$\text{if } n \equiv 0 : \omega^n = 1, f(\omega) = 1 + 1 + 1 = 3 \neq 0;$$

$$\text{if } n \equiv 1 : \omega^n = \omega, f(\omega) = \omega^2 + \omega + 1 = 0;$$

$$\text{if } n \equiv 2 : \omega^n = \omega^2, f(\omega) = \omega^4 + \omega^2 + 1 = \omega + \omega^2 + 1 = 0.$$

In the last line  $\omega^4 = \omega$ . Thus  $f(\omega) = 0$  precisely when  $3 \nmid n$ , and the same holds at  $\bar{\omega}$  by conjugation. Therefore  $x^2 + x + 1 \mid x^{2n} + x^n + 1$  if and only if  $3 \nmid n$ .

**Second method.** Work modulo  $x^3 - 1$ . Since  $x^2 + x + 1 \mid x^3 - 1$ , reduce exponents mod 3:  $x^n \equiv x^{(n \bmod 3)}$  and  $x^{2n} \equiv x^{(2n \bmod 3)}$  modulo  $x^3 - 1$ , and then modulo  $x^2 + x + 1$ . For  $3 \nmid n$  the exponents  $n, 2n$  cover the two residues  $\{1, 2\}$ , so  $x^{2n} + x^n + 1 \equiv x^2 + x + 1 \equiv 0$ ; for  $3 \mid n$  it reduces to  $1 + 1 + 1 = 3 \not\equiv 0$ .

## 6. Answer: Proof.

(a) Irreducibility over  $\mathbb{Q}$ . The roots are  $\pm\sqrt{2} \pm \sqrt{3}$ ; indeed  $(\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6}$  and  $(\sqrt{2} + \sqrt{3})^4 = (5 + 2\sqrt{6})^2 = 49 + 20\sqrt{6}$ , so  $(\sqrt{2} + \sqrt{3})^4 - 10(\sqrt{2} + \sqrt{3})^2 + 1 = 49 + 20\sqrt{6} - 10(5 + 2\sqrt{6}) + 1 = 49 + 20\sqrt{6} - 50 - 20\sqrt{6} + 1 = 0$ . No rational roots (the only candidates  $\pm 1$  give  $f(\pm 1) = -8 \neq 0$ ), so any factorization is into two monic quadratics with integer coefficients (by Gauss's lemma):

$$x^4 - 10x^2 + 1 = (x^2 + ax + b)(x^2 - ax + c),$$

where the  $x^3$ -coefficient forces the linear terms to be  $\pm a$ . Comparing coefficients:  $b + c - a^2 = -10$ ,  $a(c - b) = 0$ , and  $bc = 1$ . If  $a = 0$ :  $b + c = -10$ ,  $bc = 1$ , so  $b, c$  solve  $t^2 + 10t + 1 = 0$ , giving irrational  $t = -5 \pm 2\sqrt{6}$  — not integers. If  $c = b$ : then  $b^2 = 1 \Rightarrow b = \pm 1$ , and  $2b - a^2 = -10 \Rightarrow a^2 = 2b + 10 \in \{12, 8\}$ , neither a perfect square. So no integer factorization exists;  $f$  is irreducible over  $\mathbb{Q}$ .  
(b) Reducible modulo every prime. Use the algebraic identities, valid in any field:

$$x^4 - 10x^2 + 1 = (x^2 - 1)^2 - 8x^2 = (x^2 - 2\sqrt{2}x - 1)(x^2 + 2\sqrt{2}x - 1),$$

$$x^4 - 10x^2 + 1 = (x^2 + 1)^2 - 12x^2 = (x^2 - 2\sqrt{3}x + 1)(x^2 + 2\sqrt{3}x + 1),$$

$$x^4 - 10x^2 + 1 = (x^2 - 5)^2 - 24 = (x^2 - 5 - 2\sqrt{6})(x^2 - 5 + 2\sqrt{6}).$$

These exhibit  $f$  as a difference of squares whenever 2, 3, or 6 is a quadratic residue mod  $p$ . For any odd prime  $p$ , among the three numbers 2, 3,  $6 = 2 \cdot 3$ , at least one is a quadratic residue: if 2 and 3 are both nonresidues, their product 6 is a residue (Legendre symbol is multiplicative,  $(-1)(-1) = +1$ ). So one of the three factorizations has its surd  $\sqrt{2}, \sqrt{3}, \sqrt{6}$  lying in  $\mathbb{F}_p$ , splitting  $f$  into two quadratics over  $\mathbb{F}_p$ . For  $p = 2$ ,  $f \equiv x^4 + 1 \equiv (x + 1)^4 \pmod{2}$ . Hence  $f$  is reducible modulo every prime, even though it is irreducible over  $\mathbb{Q}$ . ■

**Second method.** For (b), the cleaner phrasing:  $f$  is the minimal polynomial of  $\sqrt{2} + \sqrt{3}$ , whose splitting field is  $\mathbb{Q}(\sqrt{2}, \sqrt{3})$  with Galois group  $(\mathbb{Z}/2)^2$ . Since this group has no element of order 4, the Frobenius at any unramified prime is a product of cycles of length  $\leq 2$  — so  $f \pmod{p}$  never stays irreducible; it always splits into factors of degree  $\leq 2$ . This is the standard example that a polynomial reducible mod every prime can still be irreducible over  $\mathbb{Q}$ .

## 7. Answer: Proof.

We use the fundamental lemma: for a polynomial  $P \in \mathbb{Z}[x]$  and integers  $u \neq v$ ,  $(u - v) \mid (P(u) - P(v))$ . (Reason: each term  $c_k(u^k - v^k)$  is divisible by  $u - v$ .) Apply it to the three given equations:

$$a - b \mid P(a) - P(b) = b - c,$$

$$b - c \mid P(b) - P(c) = c - a,$$

$$c - a \mid P(c) - P(a) = a - b.$$

Taking absolute values and chaining the divisibilities,

$$|a - b| \leq |b - c| \leq |c - a| \leq |a - b|.$$

Therefore all three are equal to a common positive value  $d := |a - b| = |b - c| = |c - a|$  (positive since  $a, b, c$  are distinct). Now consider the signed differences. They always satisfy

$$(a - b) + (b - c) + (c - a) = 0.$$

But each of the three summands equals  $\pm d$ . A sum of three terms, each  $\pm d$  with  $d > 0$ , can never be zero: its value is  $d$  times an odd integer ( $\#(+d) - \#(-d) \equiv 3 \pmod{2}$ ), hence nonzero. This contradiction shows no such  $P$  and distinct  $a, b, c$  can exist. ■ (Remark: a 2-cycle  $P(a) = b$ ,  $P(b) = a$  is perfectly possible — e.g.  $P(x) = a + b - x$  — so the obstruction is genuinely about the odd length 3 of the cycle.)

**Second method.** Geometric/extremal variant: WLOG  $a$  is the largest of  $a, b, c$ . From  $|a - b| = |b - c| = |c - a| = d$  the three points are mutually equidistant on the integer line, which is impossible for three distinct collinear points (a line cannot host an equilateral triple). Equivalently, three distinct reals cannot have all pairwise distances equal, so no such configuration exists.



**8. Answer: Proof; for  $x^3 - 7x + 3$  both sides equal  $-21$ .**

Let  $e_1 = a + b + c$ ,  $e_2 = ab + bc + ca$ ,  $e_3 = abc$  be the elementary symmetric functions, so  $a, b, c$  are the roots of  $t^3 - e_1t^2 + e_2t - e_3$ . We are given  $e_1 = 0$ . Newton's identities relate power sums to the  $e_i$ : for three variables,

$$p_1 = e_1, \quad p_2 = e_1p_1 - 2e_2, \quad p_3 = e_1p_2 - e_2p_1 + 3e_3,$$

and for  $k \geq 4$  (where  $e_k = 0$  for three variables),

$$p_k = e_1p_{k-1} - e_2p_{k-2} + e_3p_{k-3}.$$

Substituting  $e_1 = 0$  collapses everything:

$$p_1 = 0, \quad p_2 = -2e_2, \quad p_3 = 3e_3,$$

$$p_4 = -e_2p_2 + e_3p_1 = -e_2(-2e_2) + 0 = 2e_2^2,$$

$$p_5 = -e_2p_3 + e_3p_2 = -e_2(3e_3) + e_3(-2e_2) = -3e_2e_3 - 2e_2e_3 = -5e_2e_3.$$

Now compute the two sides of the claimed identity in terms of  $e_2, e_3$ :

$$\frac{p_5}{5} = \frac{-5e_2e_3}{5} = -e_2e_3, \quad \frac{p_3}{3} \cdot \frac{p_2}{2} = \frac{3e_3}{3} \cdot \frac{-2e_2}{2} = e_3 \cdot (-e_2) = -e_2e_3.$$

The two expressions are identical, proving  $\frac{p_5}{5} = \frac{p_3}{3} \cdot \frac{p_2}{2}$  whenever  $a + b + c = 0$ . Check. For  $x^3 - 7x + 3$  we read  $e_1 = 0$ ,  $e_2 = -7$ ,  $e_3 = -3$ . Then  $\frac{p_5}{5} = -e_2e_3 = -(-7)(-3) = -21$ , and  $\frac{p_3}{3} \cdot \frac{p_2}{2} = e_3 \cdot (-e_2) = (-3) \cdot 7 = -21$ . Both sides equal  $-21$ , as claimed. ■

**Second method.** Avoiding Newton's identities, use  $a + b + c = 0$  directly. Squaring gives  $p_2 = -2e_2$  since  $0 = p_2 + 2e_2$ . Because each variable satisfies  $t^3 = -e_2t + e_3$  (the relation from  $t^3 - e_2t + e_3 = 0$  when  $e_1 = 0$ ), summing over  $a, b, c$ :  $p_3 = -e_2p_1 + 3e_3 = 3e_3$ . Multiplying that cubic relation by  $t^2$  gives  $t^5 = -e_2t^3 + e_3t^2$ ; summing yields  $p_5 = -e_2p_3 + e_3p_2 = -e_2(3e_3) + e_3(-2e_2) = -5e_2e_3$ . Then  $p_5/5 = -e_2e_3 = (p_3/3)(p_2/2)$ .